GENEDIS 2026 SATELLITE HACKATHON · PREDICT THE FUTURE

MOSAIC-3

Multi-Omics Stability, Integration and Clinical Insight: a reproducible, disease-agnostic framework to quantify predictive contribution, biomarker stability, and exploratory cross-indication molecular overlap in NSCLC and rheumatoid arthritis.

LUAD / LUSC survival

RA vs OA classification

RNA · CNV · HM450 · ATAC · ChIP

Nested / repeated CV

Executive summary

440

LUAD complete cases

360

LUSC complete cases

20

RA/OA complete donors

4

RA molecular layers

MOSAIC-3 evaluates two linked questions: (1) how much each molecular layer adds to clinical or transcriptomic baselines, and (2) how reproducibly models prioritize genes and pathways when the training samples and available layers change. A cross-indication comparison is used to generate biological hypotheses, not to imply shared causation or cellular equivalence.

> **Scientific interpretation.** All findings are exploratory. The archived NSCLC metrics come from one five-fold outer-CV repetition; the RA metrics come from two five-fold repetitions on only 20 donors. Neither benchmark constitutes independent external validation or a clinical prediction tool.

1. Cohorts and data provenance

NSCLC: TCGA-LUAD / TCGA-LUSC

Public UCSC Xena cohort matrices were harmonized using primary tumour samples (sample type 01), one deterministic sample per case, complete RNA/CNV/HM450 profiles, and valid overall-survival time/event fields.

COHORT	N_MODEL_COMPLETE	EVENTS	CENSORED	RNA_FEATURES	CNV_FEATURES	METHYLATION_FEAT
LUAD	440.000	157.000	283.000	20530.000	24776.000	396065.000
LUSC	360.000	154.000	206.000	20530.000	24776.000	396065.000

Rheumatoid arthritis: synovial tissue

GEO accessions: GSE112656 (RNA-seq), GSE112654 (ATAC-seq), and GSE112655 (ChIP-seq). The benchmark uses 10 RA and 10 OA donors with four aligned layers: RNA, ATAC, H3K27ac and H3K4me3. WGBS and raw FASTQ/SRA are excluded.

- RNA features: 57,763
- ATAC features: 17,939
- H3K27ac features: 17,102
- H3K4me3 features: 16,290

MC3 gene-level SNV files are acquired as an exploratory layer but excluded from the primary benchmark until callability/coverage can be established. No controlled-access data are required.

2. Methodology

NSCLC survival

- Clinical-only baseline with age, sex and stage where available.
- Cox Elastic Net with nested tuning and repeated outer folds.
- RNA, CNV and HM450 are assessed individually and in combinations.
- Random Survival Forest is included as a sensitivity comparison.
- Harrell's C-index is reported by model and outer fold.

RA classification

- Donor-level, stratified outer folds; each donor stays in one test fold per repetition.
- Elastic Net logistic regression with inner-fold lambda selection.
- Single-layer and multi-layer ablations are compared with RNA-only baseline.

- Linear SVM is an additional sensitivity analysis.
- Metrics: ROC-AUC, balanced accuracy, sensitivity, specificity, MCC and Brier score.

Leakage controls. NSCLC imputation, variance filtering, scaling and clinical encoding are estimated from training partitions; the endpoint fields are excluded from predictors. RA supervised feature selection, imputation, variance ranking and scaling are fitted within training folds. The archived RA run calculated label-blind TMM/logCPM normalization once before cross-validation; this is disclosed as a transductive preprocessing limitation, not as label-guided selection. The current source retains this documented behaviour to preserve the pipeline and includes a modality-name fix for future runs.

3. NSCLC survival results

Mean Harrell C-index and fold standard deviation for LUAD and LUSC.

Mean Harrell C-index and fold standard deviation for LUAD and LUSC.

LUAD model metrics

MODEL	MEAN_C_INDEX	SD_C_INDEX	FOLDS
CLINICAL	0.654	0.067	5.000
CLINICAL_ALL	0.592	0.070	5.000
CLINICAL_CNV	0.632	0.042	5.000
CLINICAL_METHYLATION	0.613	0.023	5.000
CLINICAL_RNA	0.627	0.104	5.000
CNV	0.488	0.027	5.000
METHYLATION	0.589	0.053	5.000

MODEL	MEAN_C_INDEX	SD_C_INDEX	FOLDS
OMICS_ALL	0.609	0.071	5.000
RNA	0.609	0.097	5.000
RSF_CLINICAL	0.680	0.070	5.000
RSF_CLINICAL_ALL	0.575	0.075	5.000

LUSC model metrics

MODEL	MEAN_C_INDEX	SD_C_INDEX	FOLDS
CLINICAL	0.504	0.075	5.000
CLINICAL_ALL	0.537	0.085	5.000
CLINICAL_CNV	0.538	0.076	5.000
CLINICAL_METHYLATION	0.509	0.064	5.000
CLINICAL_RNA	0.515	0.052	5.000
CNV	0.541	0.079	5.000
METHYLATION	0.525	0.082	5.000
OMICS_ALL	0.529	0.102	5.000
RNA	0.520	0.049	5.000
RSF_CLINICAL	0.565	0.093	5.000
RSF_CLINICAL_ALL	0.541	0.054	5.000

Incremental C-index relative to the Cox clinical-only baseline; positive values indicate a higher observed C-index.

Incremental C-index relative to the Cox clinical-only baseline; positive values indicate a higher observed C-index.

LUAD. RSF with clinical predictors achieved the highest archived mean C-index (0.680, SD 0.070), compared with the Cox clinical baseline (0.654, SD 0.067). Among the Cox modality combinations, none exceeded the clinical-only baseline in this run; clinical + CNV reached 0.632.

LUSC. RSF with clinical predictors achieved 0.565 (SD 0.093), compared with the Cox clinical baseline at 0.504 (SD 0.075). The highest Cox multi-omic combination was clinical + CNV at 0.538; the full clinical + all-omics model reached 0.537.

> **Do not over-interpret fold maxima.** The LUAD and LUSC figures above are fold averages across five folds from one outer repetition. They are not repeated-CV estimates. Individual fold maxima are not presented as the headline score.

4. RA multi-omics classification

Mean out-of-fold ROC-AUC and balanced accuracy across two outer repetitions.

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MODEL	N_REPETITIONS	MEAN_AUC	SD_AUC	MEAN_BALANCED_ACCURACY	SD_BALANCED_ACCURACY	I
ALL	2.000	1.000	0.000	0.975	0.035	
ALL_SVM	2.000	1.000	0.000	1.000	0.000	1
ATAC	2.000	0.680	0.057	0.550	0.000	
ATAC+CHIP	2.000	0.585	0.120	0.550	0.071	
H3K27ac	2.000	0.765	0.049	0.675	0.106	
H3K4me3	2.000	0.460	0.057	0.475	0.035	
RNA	2.000	1.000	0.000	0.975	0.035	
RNA_SVM	2.000	1.000	0.000	1.000	0.000	1
RNA+ATAC	2.000	1.000	0.000	0.975	0.035	
RNA+CHIP	2.000	1.000	0.000	0.975	0.035	

RNA-only and several RNA-containing configurations achieved a mean AUC of 1.000 across the two archived repetitions; RNA-only and the ALL multi-omics model had mean balanced accuracy of 0.975. The SVM sensitivity variants reached 1.000 for both reported metrics, but do not provide a Brier score. ATAC-only and H3K4me3-only performance was lower and less balanced. These unusually high values are highly susceptible to overfitting and sampling variation with n=20 and must be treated as exploratory.

5. Biomarker stability

Mean pairwise Jaccard similarity between outer-fold selected feature sets for RA models.

Mean pairwise Jaccard similarity between outer-fold selected feature sets for RA models.

MODEL	COMPARISONS	MEAN_JACCARD	SD_JACCARD
RNA	45.000	0.398	0.145
ATAC	44.000	0.130	0.135
H3K27ac	45.000	0.207	0.144
H3K4me3	17.000	0.011	0.045
RNA+ATAC	45.000	0.339	0.118
RNA+CHIP	45.000	0.388	0.128
ATAC+CHIP	39.000	0.069	0.101
ALL	45.000	0.330	0.110

RNA-only had mean Jaccard similarity 0.398 (45 pairwise comparisons), followed by RNA+CHIP at 0.388 and RNA+ATAC at 0.339. H3K4me3-only had the lowest reported stability (0.011, 17 comparisons). The archived RA feature table was audited to remove duplicate feature selections within the same model/repetition/fold, so selection frequencies no longer exceed 1.0. Because archived feature identifiers omitted modality prefixes, exact historical per-layer attribution cannot be reconstructed; the updated R source now preserves modality-qualified column names in future runs.

6. Cross-indication comparison and biological grounding

RA_STABLE_GENES	LUAD_STABLE_GENES	LUSC_STABLE_GENES	SHARED_RA_LUAD	SHARED_RA_LUSC	NOTE
63.000	503.000	38.000	3.000	0.000	Explor

Three mapped gene symbols were shared between RA and LUAD stable-feature sets: NR0B1, PAX9 and SPINK1. No mapped shared genes were reported for RA and LUSC in the archived comparison.

GENE_SYMBOL	CONTEXT	RA_SELECTION_FREQUENCY	NSCLC_SELECTION_FREQUENCY
NR0B1	RA_LUAD_shared	0.100	0.400
PAX9	RA_LUAD_shared	0.100	0.200
SPINK1	RA_LUAD_shared	0.100	0.200

Top GO Biological Process terms for the RA stable-gene set, ranked by adjusted p-value.

Top GO Biological Process terms for the RA stable-gene set, ranked by adjusted p-value.

GO	DESCRIPTION	COUNT	PVALUE	PADJ
GO:0007204	positive regulation of cytosolic calcium ion concentration	5.000	0.000	0.001
GO:0070098	chemokine-mediated signaling pathway	3.000	0.001	0.004
GO:0050679	positive regulation of epithelial cell proliferation	3.000	0.001	0.004
GO:0060326	cell chemotaxis	3.000	0.001	0.005
GO:0006874	intracellular calcium ion homeostasis	3.000	0.003	0.011
GO:0006935	chemotaxis	3.000	0.004	0.011
GO:0019722	calcium-mediated signaling	3.000	0.004	0.011
GO:0070588	calcium ion transmembrane transport	3.000	0.005	0.012
GO:0007155	cell adhesion	5.000	0.011	0.021

GO	DESCRIPTION	COUNT	PVALUE	PADJ
GO:0008284	positive regulation of cell population proliferation	5.000	0.013	0.023
GO:0030154	cell differentiation	4.000	0.070	0.115
GO:0006954	inflammatory response	3.000	0.094	0.141
GO:0006955	immune response	3.000	0.119	0.165
GO:0043066	negative regulation of apoptotic process	3.000	0.147	0.189
GO:0045892	negative regulation of DNA-templated transcription	3.000	0.202	0.242
GO:0045944	positive regulation of transcription by RNA polymerase II	5.000	0.245	0.275
GO:0007165	signal transduction	4.000	0.405	0.421
GO:0006357	regulation of transcription by RNA polymerase II	5.000	0.421	0.421

GO Biological Process results include chemokine-mediated signalling, chemotaxis, cell adhesion and calcium-related processes among the adjusted-significant terms. This is an exploratory over-representation analysis; it does not demonstrate disease mechanism, causal direction, or cell-state equivalence between synovial fibroblasts and cancer-associated fibroblasts.

7. Computational budget and reproducibility

The archived benchmark used a reduced execution budget because of the submission time window and available laptop compute capacity: NSCLC used one outer repetition with five folds, two inner folds, up to 250 features per layer and eight lambda values; RA used two outer repetitions with five folds, three inner folds and up to 120 features per layer. The repository configuration is set to a fuller jury rerun with three outer repetitions for both NSCLC and RA, five outer folds, three inner folds and the higher feature/lambda settings defined in the configuration. This fuller run is expected to take substantially longer and may require more memory; the archived metrics in this report will not change unless the benchmark is rerun.

The pipeline is fresh-start capable: it downloads public TCGA/Xena and GEO inputs, prepares aligned sample/donor matrices, runs preflight QC, benchmarks the models, performs cross-indication enrichment and generates the report. Raw data and the large processed input matrices are not packaged in the repository; they are re-created by the

download scripts. Versioned result tables, model predictions, feature-selection outputs, QC summaries and this report are included.

8. Limitations

- NSCLC estimates are based on one outer repetition (five folds) and therefore have uncertain run-to-run variability.
- RA has only 20 complete donors. Inner training partitions can contain fewer than eight observations in a class; glmnet warnings were recorded in the warning CSV.
- RA RNA TMM/logCPM was calculated once before outer cross-validation using no labels. This is label-blind but cohort-wide preprocessing and may be considered transductive; a strict confirmatory analysis should estimate normalization reference factors within each training split.
- Archived RA feature identifiers lacked modality prefixes; the report corrects gene-level duplicate counting, but exact legacy layer attribution is unavailable. The current R source fixes feature naming for new runs.
- The archived NSCLC console output recorded glmnet numerical/convergence warnings at highly regularised lambda settings; a complete per-fold NSCLC warning file was not retained. The RA warning records are included in results/RA/ra_model_warnings.csv. Review these caveats before interpreting individual models.
- GEO annotation produced one-to-many mapping warnings, and 403 genes were excluded by the promoter/annotation mapping step in the preparation log.
- No independent external validation cohort is reported here. No clinical utility, treatment response, prognosis deployment or diagnostic validity is claimed.
- Cross-indication overlap is exploratory and subject to identifier mapping, feature-selection and assay differences.

9. Hackathon deliverables and compliance

- Public sources and accession IDs documented; no new controlled-access approval is needed.
- Supervised endpoints are explicit: overall survival time/event for NSCLC and RA versus OA for RA.
- Baselines are included: clinical-only for NSCLC and RNA-only for RA.

- Leakage controls are applied to model fitting; the RA label-blind normalization caveat is explicitly disclosed.
- Source code, fresh-start download scripts, QC, model outputs, provenance notes, and limitations are included.
- Raw references and large matrices are excluded from the Git repository; they can be reconstructed through the pipeline.

Official challenge page: GeNeDis 2026 Satellite Hackathon - Predict the Future.
